Lecture 10

Conjugate priors: Part 2

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(Edited from the lecture notes by Brian J. Reich, NC State)

Conjugate priors: Recap

- A prior is called **conjugate** if the posterior belongs to the same parametric family as that of the prior.
- The advantage of a conjugate prior is that the posterior has a closed form expression.
- We have seen that in case of a binomial or negative binomial likelihood and a beta prior, the posterior is also a beta distribution.
- In case of a multinomial likelihood and a Dirichlet prior, the posterior distribution is also Dirichlet.

Estimating rate/intensity using the Poisson/gamma model

- Estimating a rate has many applications:
  - Number of IED explosions per day in Afghanistan
  - Number of Covid 19 cases per day at IIT-K campus
  - Number of diseased trees per square mile in a forest
- Let $\lambda > 0$ be the rate we are trying to estimate.

- We make observations over a period (or region) of length (or area) N and observe $Y \in \{0, 1, 2, ...\}$ events.

- The expected number of events is $N\lambda$ so that λ is the expected number of events per unit time.

- We would like obtain the posterior of λ, a 95% interval, and a test that λ equals some predetermined value λ_0.

Frequentist/Bayesian analysis - Likelihood

- Since Y is a count with mean $N\lambda$, a natural model is

$$Y|\lambda \sim \text{Poisson}(N\lambda).$$

- PMF: $P(Y = y|\lambda) = \frac{\exp(-N\lambda)(N\lambda)^y}{y!}$.

- Mean: $\text{E}(Y|\lambda) = N\lambda$.

- Variance: $\text{V}(Y|\lambda) = N\lambda$.

Frequentist analysis

- The maximum likelihood estimate is the sample rate

$$\hat{\lambda} = Y/N.$$

- For large N and Y, the sampling distribution of $\hat{\lambda}$ is approximately

$$\hat{\lambda} \sim \text{Normal}\left(\lambda, \frac{\lambda}{N}\right).$$

- The standard error (standard deviation of the sampling distribution) is approximated as

$$\text{SE}(\hat{\lambda}) \approx \sqrt{\frac{\hat{\lambda}}{N}}.$$

- A 95% CI (for large sample case) is then

$$\hat{\lambda} \pm 2\text{SE}(\hat{\lambda}).$$

Bayesian analysis - Prior

- The parameter λ is continuous and positive, therefore a natural prior is
$$\lambda \sim \text{Gamma}(a, b).$$
- PDF: $f(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda)$.
- Mean: $\text{E}(\lambda) = \frac{a}{b}$.
- Variance: $\text{V}(\lambda) = \frac{a}{b^2}$.

Derivation of the posterior

- The posterior is

$$\begin{aligned} p(\lambda|Y) &\propto f(Y|\lambda)\pi(\lambda) \\ &\propto \exp(-N\lambda)\lambda^{Y}\lambda^{a-1}\exp(-b\lambda) \\ &\propto \exp(-[N+b]\lambda)\lambda^{Y+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is
$$\lambda|Y \sim \text{Gamma}(a+Y, b+N).$$
- Our prior choice for λ was gamma and the posterior we obtain is again gamma.
- Thus, Gamma(a, b) is a conjugate prior for λ.

Shrinkage

- The posterior mean is
$$\hat{\lambda}_B = \mathrm{E}(\lambda | Y) = \frac{Y + a}{N + b}.$$
- The posterior mean is between the sample rate Y/N and the prior mean a/b:
$$\hat{\lambda}_B = w\frac{Y}{N} + (1 - w)\frac{a}{b},$$
where the weight on the sample rate is $w = \frac{N}{N+b}$.
- $\hat{\lambda}_B$ close to Y/N, when b is negligible in comparison of N.
- $\hat{\lambda}_B$ shrinks towards the prior mean a/b, when N/b is negligible.

Selecting the prior

- The posterior is $\lambda | Y \sim \text{Gamma}(a + Y, b + N)$.
- Therefore, a and b can be interpreted as the "prior number of events and observation time".
- This is useful for specifying the prior.
- If we have no information about λ before collecting data, we should select a and b to be tiny; for example, $a = 0.1, b = 0.1$.
- If historical data/expert opinion indicates that λ is likely between 0.6 and 0.8, we should select a and b in such a way that $\text{Prob}(0.6 < \lambda < 0.8)$ is high.

Posterior with two observations

- Derive the posterior if $Y_1 \sim \text{Poisson}(N_1\lambda)$; $Y_2 \sim \text{Poisson}(N_2\lambda)$; and $\lambda \sim \text{Gamma}(a, b)$. If not mentioned, assume Y_1 and Y_2 to be independent.

- The posterior is

$$\begin{aligned} p(\lambda|Y_1, Y_2) &\propto f(Y_1, Y_2|\lambda)\pi(\lambda) \\ &\propto f(Y_1|\lambda)f(Y_2|\lambda)\pi(\lambda) \\ &\propto \exp(-N_1\lambda)\lambda^{Y_1}\exp(-N_2\lambda)\lambda^{Y_2}\lambda^{a-1}\exp(-b\lambda) \\ &\propto \exp(-[N_1 + N_2 + b]\lambda)\lambda^{Y_1+Y_2+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\lambda|Y_1, Y_2 \sim \text{Gamma}(a + Y_1 + Y_2, b + N_1 + N_2).$$

Posterior with two observations

- Derive the posterior if $Y_1 \sim \text{Poisson}(N_1\lambda)$; $Y_2 \sim \text{Poisson}(N_2\lambda)$; and $\lambda \sim \text{Gamma}(a, b)$. If not mentioned, assume Y_1 and Y_2 to be independent.

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$$\begin{aligned}
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&\propto f(Y_1|\lambda)f(Y_2|\lambda)\pi(\lambda) \\
&\propto \exp(-N_1\lambda)\lambda^{Y_1}\exp(-N_2\lambda)\lambda^{Y_2}\lambda^{a-1}\exp(-b\lambda) \\
&\propto \exp(-[N_1 + N_2 + b]\lambda)\lambda^{Y_1+Y_2+a-1}.
\end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is
$$\lambda|Y_1, Y_2 \sim \text{Gamma}(a + Y_1 + Y_2, b + N_1 + N_2).$$

Posterior with m observations

- Derive the posterior if $Y_1, ..., Y_m \sim \text{Poisson}(N\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.
- The posterior is

$$\begin{aligned} p(\lambda|Y_1,\ldots,Y_m) &\propto f(Y_1,\ldots,Y_m|\lambda)\pi(\lambda) \\ &\propto f(Y_1|\lambda)\times\ldots\times f(Y_m|\lambda)\pi(\lambda) \\ &\propto \exp(-N\lambda)\lambda^{Y_1}\times\ldots\times\exp(-N\lambda)\lambda^{Y_m}\lambda^{a-1}\exp(-b\lambda) \\ &\propto \exp(-[mN+b]\lambda)\lambda^{\sum_{i=1}^m Y_i+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\lambda|Y_1,\ldots,Y_m \sim \text{Gamma}\left(a+\sum_{i=1}^m Y_i, b+mN\right).$$

Posterior with m observations

- Derive the posterior if $Y_1, ..., Y_m \sim \text{Poisson}(N\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.
- The posterior is

$$\begin{aligned} p(\lambda | Y_1, \ldots, Y_m) &\propto f(Y_1, \ldots, Y_m | \lambda)\pi(\lambda) \\ &\propto f(Y_1|\lambda) \times \ldots \times f(Y_m|\lambda)\pi(\lambda) \\ &\propto \exp(-N\lambda)\lambda^{Y_1} \times \ldots \times \exp(-N\lambda)\lambda^{Y_m}\lambda^{a-1}\exp(-b\lambda) \\ &\propto \exp(-[mN+b]\lambda)\lambda^{\sum_{i=1}^m Y_i + a - 1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\lambda | Y_1, \ldots, Y_m \sim \text{Gamma}\left(a + \sum_{i=1}^m Y_i, b + mN\right).$$

AB testing example

- A tech company runs their regular user interface for $N_1 = 8$ hours and gets $Y_1 = 4721$ clicks.
- The next day they launch a new user interface for $N_2 = 8$ hours and get $Y_2 = 5209$ clicks.
- Assuming non-informative conjugate priors, determine if the new user interface has a higher click rate.

AB testing example

- Period 1: the likelihood is $Y_1|\lambda_1 \sim \text{Poisson}(N_1\lambda_1)$.
- The conjugate prior is $\lambda_1 \sim \text{Gamma}(a, b)$.
- The posterior is $\lambda_1|Y_1 \sim \text{Gamma}(Y_1 + a, N_1 + b)$.
- Period 2: Similarly, $\lambda_2|Y_2 \sim \text{Gamma}(Y_2 + a, N_2 + b)$.

Monte Carlo approximation

```
> S <- 100000
> a <- b <- 0.1
> N1 <- N2 <- 8
> Y1 <- 4721
> Y2 <- 5209
>
> # MC samples
> lambda1 <- rgamma(S,Y1+a,N1+b)
> lambda2 <- rgamma(S,Y2+a,N2+b)
>
> # Prob(new interface is better|data)
> mean(lambda2>lambda1)
[1] 1
> # The new interface almost surely works!
```

Exponential models

- The first continuous distribution we will discuss is the exponential distribution, and we denote it by $Y \sim \text{Exponential}(\lambda)$ where $\lambda > 0$.
- Domain: $Y \in (0, \infty)$.
- PDF: $f(y) = \lambda \exp(-\lambda y)$.
- Mean: $\text{E}(Y|\lambda) = \frac{1}{\lambda}$.
- Variance: $\text{V}(Y|\lambda) = \frac{1}{\lambda^2}$.

Bayesian analysis - Prior

- The parameter λ is continuous and positive, therefore a natural prior is
$$\lambda \sim \text{Gamma}(a, b).$$
- PDF: $f(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda)$.
- Mean: $\text{E}(\lambda) = \frac{a}{b}$.
- Variance: $\text{V}(\lambda) = \frac{a}{b^2}$.

Derivation of the posterior

- The posterior is

$$
\begin{aligned}
p(\lambda|Y) &\propto f(Y|\lambda)\pi(\lambda) \\
&\propto \lambda \exp\left(-\lambda Y\right) \lambda^{a-1} \exp(-b\lambda) \\
&\propto \exp(-[Y+b]\lambda)\lambda^{1+a-1}.
\end{aligned}
$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is
$$\lambda|Y \sim \text{Gamma}(a+1, b+Y).$$
- Our prior choice for λ was gamma and the posterior we obtain is again gamma.
- Thus, Gamma(a, b) is a conjugate prior for λ.

Posterior with n observations

- Derive the posterior if $Y_1, ..., Y_n \sim \text{Exponential}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.
- The posterior is

$$\begin{aligned} p(\lambda|Y_1,\ldots,Y_n) &\propto f(Y_1,\ldots,Y_n|\lambda)\pi(\lambda) \\ &\propto f(Y_1|\lambda)\times\ldots\times f(Y_n|\lambda)\pi(\lambda) \\ &\propto \lambda\exp(-\lambda Y_1)\times\ldots\times\lambda\exp(-\lambda Y_n)\lambda^{a-1}\exp(-b\lambda) \\ &\propto \exp\left(-\left[\sum_{i=1}^n Y_i + b\right]\lambda\right)\lambda^{n+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\lambda|Y_1,\ldots,Y_n \sim \text{Gamma}\left(a+n, b+\sum_{i=1}^n Y_i\right).$$

Posterior with n observations

- Derive the posterior if $Y_1, ..., Y_n \sim \text{Exponential}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.
- The posterior is

$$
\begin{aligned}
p(\lambda | Y_1, \ldots, Y_n) &\propto f(Y_1, \ldots, Y_n | \lambda)\pi(\lambda) \\
&\propto f(Y_1|\lambda) \times \ldots \times f(Y_n|\lambda)\pi(\lambda) \\
&\propto \lambda \exp(-\lambda Y_1) \times \ldots \times \lambda \exp(-\lambda Y_n)\lambda^{a-1} \exp(-b\lambda) \\
&\propto \exp\left(-\left[\sum_{i=1}^{n} Y_i + b\right]\lambda\right)\lambda^{n+a-1}.
\end{aligned}
$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$
\lambda | Y_1, \ldots, Y_n \sim \text{Gamma}\left(a + n, b + \sum_{i=1}^{n} Y_i\right).
$$

Gamma models with known shape parameter

- The next continuous distribution we will discuss is the gamma distribution, and we denote it by $Y \sim \text{Gamma}(\alpha, \beta)$ where $\alpha > 0$ is known and $\beta > 0$.

- Domain: $Y \in (0, \infty)$.

- PDF: $f(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} \exp(-\beta y)$.

- Mean: $\text{E}(Y|\beta) = \frac{\alpha}{\beta}$.

- Variance: $\text{V}(Y|\beta) = \frac{\alpha}{\beta^2}$.

Bayesian analysis - Prior

- The parameter β is continuous and positive, therefore a natural prior is
$$\beta \sim \mathrm{Gamma}(a, b).$$
- PDF: $f(\beta) = \frac{b^a}{\Gamma(a)} \beta^{a-1} \exp(-b\beta)$.
- Mean: $\mathrm{E}(\beta) = \frac{a}{b}$.
- Variance: $\mathrm{V}(\beta) = \frac{a}{b^2}$.

Derivation of the posterior

- The posterior is

$$\begin{aligned} p(\beta|Y) &\propto f(Y|\beta)\pi(\beta) \\ &\propto \beta^{\alpha}\exp(-\beta Y)\beta^{a-1}\exp(-b\beta) \\ &\propto \exp(-[Y+b]\beta)\beta^{\alpha+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is
$$\beta|Y \sim \text{Gamma}(a+\alpha, b+Y).$$
- Our prior choice for β was gamma and the posterior we obtain is again gamma.
- Thus, Gamma(a, b) is a conjugate prior for β.

Posterior with n observations

- Derive the posterior if $Y_1, ..., Y_n \sim \text{Gamma}(\alpha, \beta)$ and $\beta \sim \text{Gamma}(a, b)$.
- The posterior is

$$\begin{aligned} p(\beta|Y_1,\ldots,Y_n) &\propto f(Y_1,\ldots,Y_n|\beta)\pi(\beta) \\ &\propto f(Y_1|\beta)\times\ldots\times f(Y_n|\beta)\pi(\beta) \\ &\propto \beta^{\alpha}\exp(-\beta Y_1)\times\ldots\times\beta^{\alpha}\exp(-\beta Y_n)\beta^{a-1}\exp(-b\beta) \\ &\propto \exp\left(-\left[\sum_{i=1}^{n} Y_i + b\right]\beta\right)\beta^{n\alpha+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\beta|Y_1,\ldots,Y_n \sim \text{Gamma}\left(a+n\alpha, b+\sum_{i=1}^{n} Y_i\right).$$

Posterior with n observations

- Derive the posterior if $Y_1, ..., Y_n \sim \text{Gamma}(\alpha, \beta)$ and $\beta \sim \text{Gamma}(a, b)$.
- The posterior is

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\begin{aligned}
p(\beta|Y_1, \ldots, Y_n) &\propto f(Y_1, \ldots, Y_n|\beta)\pi(\beta) \\
&\propto f(Y_1|\beta) \times \ldots \times f(Y_n|\beta)\pi(\beta) \\
&\propto \beta^{\alpha} \exp(-\beta Y_1) \times \ldots \times \beta^{\alpha} \exp(-\beta Y_n)\beta^{a-1} \exp(-b\beta) \\
&\propto \exp\left(-\left[\sum_{i=1}^{n} Y_i + b\right]\beta\right)\beta^{n\alpha+a-1}.
\end{aligned}
$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$
\beta|Y_1, \ldots, Y_n \sim \text{Gamma}\left(a + n\alpha, b + \sum_{i=1}^{n} Y_i\right).
$$

Thank you!